LocateAnything vs. micro-SAM for EM Segmentation

Highlight brief — internal design, EM-segmentation feasibility, and which SAM powers micro-SAM.

Compiled July 2026 · Sources: NVIDIA / Hugging Face, arXiv 2605.27365, micro-sam project docs

Bottom line up front

- **LocateAnything does not segment.** It is a vision-language *grounding & detection* model. Outputs are **bounding boxes and points only — never pixel masks**.
- **Not a drop-in for EM segmentation.** micro-SAM emits dense instance *masks* in-domain; LocateAnything emits boxes from *text prompts* on natural-image / GUI / doc data. At most an out-of-domain box-prompt front-end.
- **micro-SAM runs on SAM 1 — not SAM2 or SAM3.** Its encoders are SAM's ViT-T/B/L/H. SAM2 uses Hiera; SAM3 is Meta's Nov-2025 release. SAM2/3 are roadmap-only.

Q1 How LocateAnything works internally

Architecture: `Moon-ViT` vision encoder (native-resolution visual tokens) → MLP projector → `Qwen2.5` language decoder. ~3B params; trained on ~12M images / 138M queries / 785M boxes across detection, GUI, OCR, docs, robotics — **no microscopy data**.

Tokenization — box-aligned atomic blocks. Localization is a sequence of fixed-length blocks of four types:

Block	Role
`Semantic`	Identifies the entity / class.
`Box`	Full coordinate set `(x1,y1,x2,y2)` (or a point) as one atomic unit.
`Negative`	Signals the queried object is absent.
`End`	Termination state ending the sequence.

Parallel Box Decoding (PBD). Rather than autoregressive coordinate tokens, each box is predicted in a single parallel step — preserving geometric coherence and running **>10× faster** than Qwen3-VL. Modes: *Fast* (parallel), *Slow* (autoregressive, stable), *Hybrid* (fast with fallback).

Key point
The innovation is fast, coherent *coordinate* emission — there is no mask head and no pixel-level output.

Q2 Feasible for EM segmentation, like micro-SAM?

	micro-SAM	LocateAnything
Output	Dense instance **masks**	**Boxes + points** only
Domain	Fine-tuned on EM / LM microscopy	Natural images, GUI, OCR, docs
Prompt	Point / box / mask → mask	Text description → box

Why it's a poor fit: no masks; language-grounded prompting is awkward for dense, near-identical EM structures (organelles, membranes, vesicles); zero in-domain training.

Narrow possible role

A detection front-end in a two-stage *detect-then-segment* pipeline (boxes → SAM mask decoder) — but this needs heavy EM fine-tuning, and micro-SAM already covers automatic + interactive prompting in domain.

Verdict: **Not a replacement.** Not feasible as a standalone EM segmenter; at best an out-of-domain box-prompt generator.

Q3 Does micro-SAM use SAM2 or SAM3?

Neither — it is built on the original SAM (SAM 1), with a custom instance-segmentation decoder on top.

Generation	Image encoder	In micro-SAM?
SAM 1 (2023)	`ViT-T / B / L / H`	Shipped backbone
SAM 2 (2024)	Hiera (hierarchical)	Roadmap
SAM 3 (Nov 2025)	Meta's newest	Roadmap

How we know: micro-SAM's models — `vit_b`, `vit_l`, `vit_b_em_organelles`, etc. — are SAM's ViT encoders. SAM 2 replaced ViT with **Hiera**, so the naming alone pins micro-SAM to SAM 1. (The live release file couldn't be fetched at compile time, but every signal points to SAM 1.)
